

Tutorial 4 - Pointers

1. What is a pointer? How do you declare a pointer variable?
2.
 - (a) What unfortunate misinterpretation can occur with the following declaration?
`int* int_ptr1, int_ptr2;`
 - (b) There are four different ways to declare a pointer. In part (a) above, that is one way. What are the other three ways?
3. What is wrong in the following code?
 - (a)

```
int x = 30;
int *pX = x;
cout << "x is " << x << endl;
cout << "x is " << px << endl;
```
 - (b)

```
double x = 3.0;
int *pX = &x;
```
4. Give at least two uses of the * operator. State what the * is doing and name the use of the * that you present.
5. What is the output produced by the following code?

```
int *p1, *p2;
p1 = new int;
p2 = new int;

*p1 = 10;
*p2 = 20;
cout << *p1 << " " << *p2 << endl;

p1 = p2;
cout << *p1 << " " << *p2 << endl;

*p1 = 30;
cout << *p1 << " " << *p2 << endl;
```

How would the output change if you were to replace
 `*p1 = 30;`
and with the following?
 `*p2 = 30;`

6. What will be the output produced by the following code segment?

(a)

```
int a;  
int* p;  
a = 2;  
p = &a;  
a = a + 1;  
cout << *p;
```

(b)

```
int a;  
int* p;  
a = 4;  
p = &a;  
cout << (*p) / a;
```

(c)

```
int a;  
int b;  
int* p;  
int* q;  
a = 3;  
p = &a;  
q = p;  
*q = *q + 5;  
cout << *p; // retrieve 8 from variable a
```

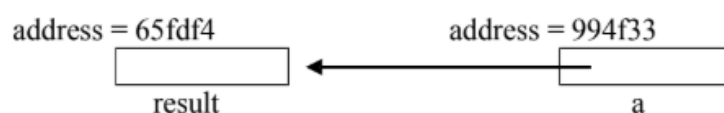
7. Given the following declarations:

```
int a = 5, b = 7;  
int *p = &a, *q = &b, *r = p;
```

What is the value for each of the following expressions? (The statements are independent of each other)

- i. `*p * a`
- ii. `(*p) * *r`
- iii. `*p + *q`
- iv. `*&*q`

8. The state of memory at a particular time is shown as follows: -



- a) Write C++ declaration statements for **a** and **result** as well as how the pointer **a** is setup to point to the variable **result** of integer type. At the same time, initialize the **result** with 10.
- b) Write the outputs of the following statements:
 - i. `cout << a;`
 - ii. `cout << *a;`
 - iii. `cout << &a;`

- iv. `cout << result;`
- v. `cout << *result;`
- vi. `cout << &result;`
- vii. `cout << *&a;`
- viii. `cout << &*a;`

9. Study the following program, determine the output of the program:-

```
void main() {
    int i = 2, j = 4, k = 6, m = 7;
    int *p = &i, *q = &j, *r = &k;

    cout << *p << endl;
    cout << *&j << endl;
    cout << *p**q**r << endl;
    cout << --*p++*q++*r << endl;
}
```

10. Show the output of the following C++ program statements: -

```
int *vPtr, *nPtr;
int v, n, r;
vPtr = &v;
nPtr = &n;
* vPtr = 5;
n = 6;

cout << "v = " << v << "\nn = " << *nPtr << endl;
r = v + *nptr;
cout << "r = " << r << endl;
(*vPtr)++;
n = * nPtr * 2;
cout << "v = " << v << "\nn = " << n << endl;
```

11. Study the following program:

```
void main( ) {
    int num1 = 100, num2 = 200, *ptr1, *ptr2;
    ptr1 = &num1;
    cout << ptr1 << endl;
    cout << *ptr1 << endl;
    ptr2 = ptr1;
    ptr1 = &num2;
    cout << ptr1 << endl;
    cout << *ptr1 << endl;
    cout << ptr2 << endl;
    cout << *ptr2 << endl;
}
```

Assuming the address of num1 is 12ff7c and the address of num2 is 12ff78, write the output of this program.

12. Consider the following C++ program, and show the output.

```
#include <iostream>
using namespace std;
void main() {
    int p1, p2;
    int v=7;
    int *ptr1, *ptr2;
    p1 = 3 * (v-2);
    ptr1 = &v;
    p2 = 3 * (++v + *ptr1);
    ptr2 = &p2;
    p2--;
    cout << "\np1 = " << p1 << "\np2 = " << p2 << "\nptr1 = " << *ptr1;
    cout << "\nptr2 = " << *ptr2 << "\nv = " << v;
}
```

13. The following code is intended to find the area and perimeter of a rectangle. However, it fails to do so. Provide the correct code to accomplish the desired result. Also, do not modify the numeric values assigned to the variables in the code. After correcting the code, what is the output?

```
double *length;
double *width;

cout << fixed << showpoint << setprecision(2);

length = new double;
length = 6.5;

&width = 3.0;

cout << "Area: " << (*length) * (&width) << ", ";
cout << "Perimeter: " << 2 * (*length + width) << endl;
```

14. Evaluate the output of the following C++ program segment:

```
int x, y, z, *ptr1, *ptr2;
ptr1 = &x;
ptr2 = &y;
*ptr1 = 24;
y = 49;
cout << "x = " << x << endl;
cout << "y = " << *ptr2 << endl;
z = x - *ptr2;
cout << "z = " << z << endl;
ptr1 = &z;
x = *ptr1 - 10;
(*ptr2)++;
cout << "x = " << x << endl;
cout << "y = " << y << endl;
```